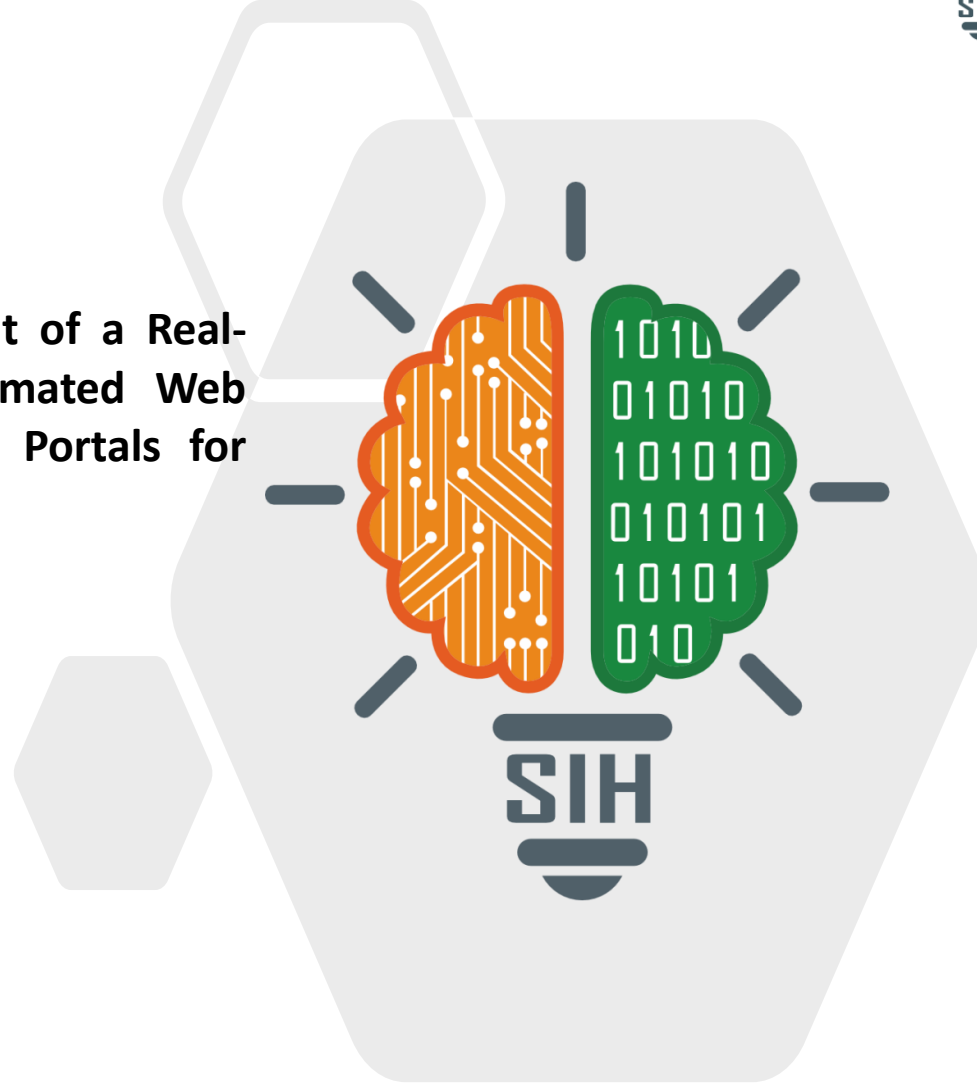


SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26056**
- **Problem Statement Title-** Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name –** Woven tech



❑ Why Problem

- **Airfare is dynamic** vary with advance booking window , festival seasons, route basis .
- **CPI** only provides state specific data that is also monthly basis .
- Currently airfare inflation is **calculated manually with only 10% of data** .



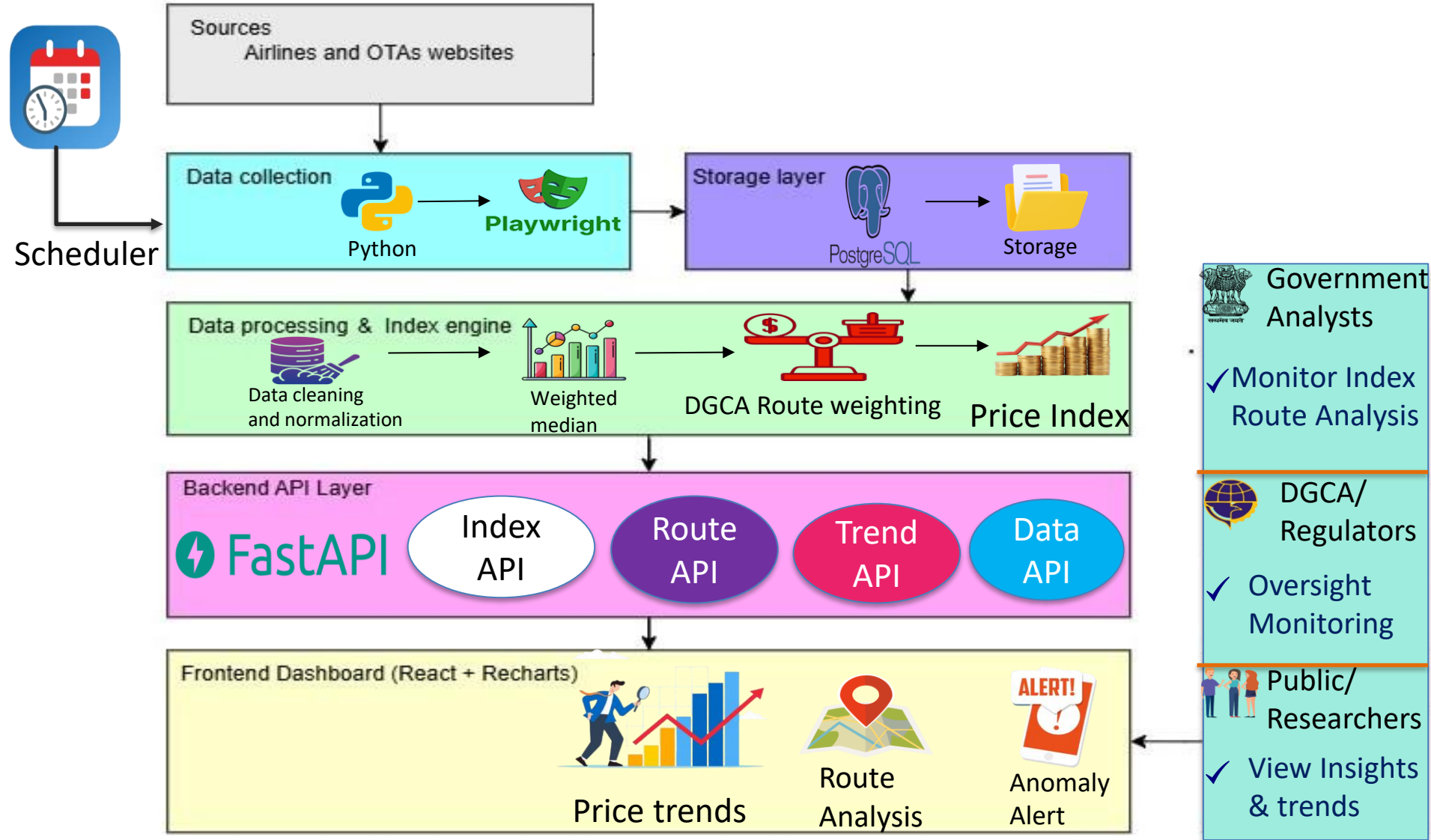
❑ Proposed Solution

- APIx frequently collects automated Real-Time airfare data **using Python + Playwright** .
- Collects data from **major airlines and OTAs** (Online Travel Aggregators) websites .
- **Gather fares from multiple Purchase windows** ex- T+1,T+7,T+15,T+30 days to observe how airfare changes with booking time.
- **Detect duplicate flights** appearing on different websites and group them together.
- **Use weighted median** method to calculate a representative fare for each flight/route.
- **Provide route specific data, price trend, show heatmap** and anomaly alerts to DGCA.

❑ Uniqueness

- Show real time and high frequency data .
- Route - level intelligence .
- Booking window analysis .
- Use DGCA traffic statistics for route prioritization .





Tech Stack



➤ Feasibility

- ✓ **Technical Feasibility** : Uses proven technologies
ex:- Python, Playwright, PostgreSQL, Fast API .
- ✓ **Economic Feasibility** : Low software cost →
Open - source technology stack .
- ✓ **User Feasibility** : Centralized airfare intelligence for
analysts .

➤ Viability

- Reduces manual data collection .
- Continuous data → Better market visibility .
- Start small → Scale progressively .
- Cost effective → open-source technology used



Risk



Anti-bot detection/ captcha blocks
scrapping .



Duplicate flight/ fare records



High data volume at scale



Mitigation



Compliant to rate limiting +
permitted/ public sources



Uses deduplication pipeline .



Scalable cloud infrastructure solves it

Impacts and benefits



Government Impact

- **Dedicated Airfare Price Index .**
- Data-driven regulatory insights .
- **Alert DGCA/regulator** to abnormal or sudden fare increases.



Economic Impact

- Route wise price intelligence .
- Identify transportation cost pressure .
- **Reduces manual analysis .**



Social Impact

- **Helps** identify high-fare periods/routes .
- **Understand** booking - time effects .
- Supports passenger – focused policies .



Technological Impact

- **Automates** high-frequency airfare data collection.
- **Integrates multiple data** sources into a unified pipeline
- **API-driven analytics dashboard .**

Primary sources :

MoSPI for CPI context - [Product | Ministry of Statistics](#)

DGCA Air Traffic Data - [Home | Directorate General of Civil Aviation | Government of India](#)

Market understanding :

Airline websites - [Book Domestic and International Flights at Lowest Airfare](#)
[SpiceJet - Flight Booking for Domestic and International, Cheap Air Tickets](#)
[Air India: Book Domestic & International Non-Stop Direct Flights](#)

OTAs websites - [Flight Booking, Cheap Flights , Air Ticket Booking at Lowest Airfare | MakeMyTrip](#)
[ixigo - Best Travel Website, Book Flights, Trains, Hotels & Buses](#)
[Flight Ticket Booking, Cheap Flights, Air Tickets at Lowest Airfare | EaseMyTrip](#)

Others :

[India Service Producer Price Index Q1 FY27: Air Travel Prices Surge, Banking Declines](#)